Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

HardwareX manuscript draft

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> **Status:** working draft generated from the repository's design files, standard operating procedure, parts list, schematics, and run logs. Bracketed `[TODO]` markers flag values or figures that still need to be confirmed or added before submission. The reproducible build and bill of materials are anchored on the lab's **current, USA-sourced induction generator** (CEIA "Power Cube" 6 kW, procured through East Coast Induction); the ~1970 LEPEL furnace is documented only as the originating prototype.

> An independent review of this draft against the HardwareX section requirements, and the resulting action checklist, are recorded in `edison-feedback/`; its highest-priority items have been folded into the BOM, specifications table, and the plan's gap checklist.

Specifications table

Property	**Value**
Hardware name	Computer-controlled, vacuum-integrated induction annealing system
Subject area	Materials science; mechanical engineering (thermal processing / grain growth)
Hardware type	Heat-treatment / annealing furnace; vacuum system; instrument control
Closest commercial analog	Vacuum induction annealing furnace (turn-key systems cost \$50k–\$200k+)
Open-source license	`[TODO: declare an OSHW-compatible license — e.g. CERN-OHL-S / MIT for code]`
Cost of hardware	`[TODO: total; retrofit + consumables ≈ $1–2k on top of the generator]`
Source file repository	`[TODO: mint a permanent DOI — e.g. Zenodo — for the repository; HardwareX does not accept a bare GitHub URL]`
Induction generator (reproducible build)	CEIA “Power Cube” 6 kW solid-state RF generator + Master Controller v3+ (East Coast Induction, USA)
Induction generator (prototype)	LEPEL high-frequency tube furnace (~1970), up to ~35 kW

Property	Value
Power-control input	Analog setpoint via LabVIEW/DAQ (4–20 mA on current controller; 0–5 mA on LEPEL)
Max sample temperature	~1400–1500 °C+ (Ni/Fe melting range)
Vacuum level	~1×10^{-6} to 1×10^{-8} Torr (Edwards T-Station 85 turbo-molecular pump)
Temperature measurement	IMPAC ISR6 dual-wavelength ratio pyrometer (range ~800–2500 °C)
Chamber	Quartz tube, KF40 optical-window flange, inert-gas vent
Crucible / susceptor	In-house machined graphite (3000 °C grade) crucible-susceptor
Work coil	Water-cooled copper, ~3″ tall, 2.5″ ID, ~6.5 turns

Hardware in context

High-temperature annealing near a metal's melting point drives controlled **grain growth**, which is central to studying microstructure–property relationships in metals such as nickel and iron. Doing this without oxidizing the sample requires both high temperature (~1400–1500 °C) and high vacuum (or inert atmosphere). Commercial vacuum induction annealing furnaces that meet these requirements are expensive turn-key systems, which puts them out of reach for many academic labs.

This work takes the opposite approach: rather than building or buying a complete furnace, we **retrofit and modernize a bare RF induction generator** into a full annealing system by adding (1) computer power control, (2) closed-loop optical temperature control, (3) a high-vacuum quartz-tube chamber, and (4) a machined graphite crucible/susceptor. The emphasis of the contribution is the **modernization stack itself**, which is deliberately **generator-agnostic**: the same control software, vacuum chamber, pyrometer feedback, crucible, and work-coil geometry transfer from one induction generator to another. The only requirement a reader's generator must satisfy is an **analog power-control input that maps roughly linearly to output**

power (e.g. 4–20 mA or 0–5 V / 0–5 mA). That portability is the reproducible result and is what makes the design reusable by other labs that already own, or can buy, such an induction generator.

A second feature that broadens the system's applicability is the **machined graphite crucible/susceptor**: because graphite couples strongly to the RF field and heats the charge indirectly by conduction and radiation, the furnace is not limited to electrically conductive samples — it can also process poorly- or non-coupling materials such as YSZ ceramic. This makes the same build useful well beyond reactive-metal annealing.

The modernization was first prototyped on a ~1970 LEPEL high-frequency induction furnace owned by the BYU Mechanical Engineering department — at the time the only departmental machine able to reach the Ni/Fe melting range. A "remote control" capability was added in August 2019 to drive the generator's power from a computer instead of the front-panel knob. The lab subsequently moved the same stack onto a current, USA-sourced solid-state generator (CEIA "Power Cube" 6 kW via East Coast Induction), whose Master Controller v3+ accepts a standard analog (4–20 mA) power setpoint. An intermediate, separately-purchased import generator (CYSI GP-15A) was evaluated but did not meet the lab's control requirements — the required analog remote-control input was lost across vendor model substitutions — and was retired. The reproducible build documented here therefore targets the current USA generator, with the LEPEL retrofit retained as historical context.

Compared with commercial vacuum annealing furnaces, this approach is roughly an order of magnitude cheaper, is repairable and modifiable by the user, and is broadly reusable: any lab with an induction generator and an analog power input can reproduce the control, vacuum, and temperature-sensing additions described below.

Hardware description

The system comprises five subsystems (see the schematics in `extracted-context/schematic_induction_furnace.md` and `extracted-context/schematic_support_stand.md`):

1. **RF generator + work coil.** A solid-state induction generator (current build: CEIA Power Cube 6 kW) drives a water-cooled copper work coil. The coil carries both the oscillating current and cooling water. Coil geometry (~3″ tall, 2.5″ ID, ~6.5 turns) was tuned to the sample/crucible stack (see `extracted-context/order_corrections_coils.md`).

2. **Vacuum + gas.** A quartz tube forms the chamber and is connected through a KF40 flange and flexible bellows to an Edwards T-Station 85 turbo-molecular pump ($\sim 1\times10^{-6}$–10^{-8} Torr). A manual vent valve and an inert-gas regulator allow controlled venting/backfilling to suppress oxidation of reactive metals.

3. **Optical temperature sensing.** A dual-wavelength (ratio) pyrometer (IMPAC ISR6; also a LumaSens 800–2500 °C unit) views the sample through a KF40 optical window, providing a non-contact temperature signal that is robust to emissivity changes and to the RF field that would corrupt thermocouples.

4. **Control / DAQ + LabVIEW.** A LabVIEW DAQ analog output sets generator power; the pyrometer signal is read back for closed-loop ramp/soak control. The VIs implement manual control, automated ramping, PID tuning, and email alerts (see `../code/`).

5. **Mechanical support stand + crucible.** A bolted support stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) positions the chamber, coil, pyrometer holder, and quartz disc. An **in-house machined graphite crucible** holds the sample and acts as the RF susceptor.

Custom-machined graphite crucible / susceptor

Because some charges of interest couple poorly to the RF field (e.g. YSZ ceramic) or must be physically contained, the sample is held in a **graphite crucible machined in-house from 3000 °C-grade graphite stock** (Cotronics 56L-3). The graphite couples strongly to the induction field and acts as a **susceptor**, transferring heat to the contained charge by conduction/radiation; this makes the same furnace usable for conductive metals (direct coupling) and for non/poorly-coupling materials (indirect, susceptor-driven heating). Crucibles are machined to fit inside the quartz tube and seat on a ceramic-cylinder / Teflon spacer stack. Cracked crucibles are repaired with a 3000 °C graphite cement (Resbond 931) rather than discarded. `[TODO: add the crucible machining drawing and photos to paper/figures/.]`

Design files summary

Design file	File type	Open source license	Location of the file
LabVIEW control VIs (furnace control, manual control, PID tuning v1/v2, email alert)	LabVIEW .vi	[TODO]	../code/induction-furnace-control-code/
Manual ramping v5 VIs	LabVIEW .vi	[TODO]	../code/manual-ramping-v5/
plotheatcurve.m (heat-curve plotter)	MATLAB	[TODO]	../code/plotheatcurve.m
Work-coil drawing	PDF / OXPS	[TODO]	docs/coils-drawing.pdf, docs/coils.oxps
System schematics	PPTX (+ extracted text/ figures)	[TODO]	docs/*.pptx, extracted-context/
Temperature-control wiring	PNG	[TODO]	docs/temp-control-modification/0-5V PLC wire design.png
KF40 overpressure centering ring	STEP	[TODO]	docs/KF Supplies/ KF40_overpressureCenteringRing.step
Graphite crucible/ susceptor machining drawing	[TODO: add]	[TODO]	[TODO]

[TODO: export the LabVIEW .vi block diagrams to PDF/PNG so non-LabVIEW readers can inspect the control logic.]

Bill of materials summary

HardwareX requires a single 7-column BOM table; groupings are denoted by the designator prefix: **GEN** = induction generator + controller (the current USA reproducible build; the legacy LEPEL is the prototype alternative, not a line item), **RETRO** = the reproducible retrofit (control / vacuum / sensing) parts, and **CONS** = consumables. Full vendor/price detail for the RETRO and CONS parts is in `extracted-context/parts_list.md` (extracted from `docs/induction_parts_list.xlsx`).

Designator	Component	Number	Cost per unit	Total cost	Source of materials	Material type
GEN-1	CEIA “Power Cube” 6 kW solid-state RF generator	1	`[TODO]`	`[TODO]`	East Coast Induction (USA)	—
GEN-2	Master Controller v3+	1	`[TODO]`	`[TODO]`	East Coast Induction (USA)	—
GEN-3	Recirculating water chiller	1	`[TODO]`	`[TODO]`	`[TODO]`	—
RETRO-1	LabVIEW-compatible DAQ (analog out + in), e.g. NI USB-6009 `[confirm model]`	1	`[TODO]`	`[TODO]`	National Instruments	—
RETRO-2	Dual-wavelength pyrometer (IMPAC ISR6 / LumaSens)	1	~$241 (used)	~$241	eBay / OEM	—
RETRO-3	Edwards T-Station 85 turbo-molecular pump	1	`[TODO]`	`[TODO]`	Edwards	—
RETRO-4	Edwards TAV5 vent valve	1	~$59	~$59	eBay	—
RETRO-5	Inert-gas regulator (Fisher FS-50)	1	~$38	~$38	Fisher Scientific	—
RETRO-6	KF40 flanges, clamps, centering	1 set	see parts list	`[TODO]`	IdealVac / McMaster	aluminum quartz

Designator	Component	Number	Cost per unit	Total cost	Source of materials	Material type
	rings, optical window					
CONS-1	Quartz tube (35 mm ID), cut to length	1–2	~$67	~$134	QSI Quartz	fused quartz
CONS-2	Graphite stock for machined crucible/susceptor (56L-3, 3000 °C)	1	~$99	~$99	Cotronics	graphite
CONS-3	Graphite repair cement (Resbond 931)	1	~$108	~$108	Cotronics	graphite
CONS-4	Torr-Seal epoxy / O-rings	1 set	see parts list	`[TODO]`	—	epoxy / elastomer

`[TODO: confirm GEN-1/2/3 and RETRO-1/3 pricing; add currency (USD) and quote year.]`

Build instructions

1. **Support stand.** Assemble the bolted stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) and mount the generator/heating head and pyrometer holder.
2. **Work coil.** Form/water-cool the copper coil to the corrected geometry (~3″ tall, 2.5″ ID, ~6.5 turns); connect to the generator head and the cooling-water circuit. Use the before/after figures from `induction_order_corrections.pptx` as the reference.
3. **Graphite crucible/susceptor.** Machine the crucible from 3000 °C graphite stock to fit inside the quartz tube and accept the sample. Seat it on the ceramic-cylinder / Teflon spacer stack. Repair any cracks with graphite cement.
4. **Vacuum chamber.** Mount the quartz tube with the KF40 flange and optical window (set the window with the overpressure centering ring + clamp, or Torr-Seal). Connect the flexible bellows to the Edwards T-Station 85 turbo pump; install the manual vent valve and inert-gas line.

5. **Control wiring.** Wire the DAQ analog output to the generator's analog power-control input (4–20 mA on the current controller; 0–5 mA on the LEPEL prototype). Wire the pyrometer output into the DAQ analog input and, for closed-loop ramping, the pyrometer/PLC temperature-control path (`docs/temp-control-modification/`).
6. **Software.** Deploy the LabVIEW VIs from `../code/` and configure the DAQ channels; verify manual control before enabling automated ramp/soak.

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[TODO: curate a clean, photographed build sequence — candidate
photos exist in docs/data_log/.../IFrun036_* and the PPTX media.]
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Operation instructions

Condensed from the lab SOP (`extracted-context/SOP_induction_200901.md`):

1. **Load the sample.** Vent the chamber (confirm atmospheric pressure; never open the vent valve while the turbo pump is spinning). Stack sample → optional alumina plate → ceramic cylinder → Teflon spacer inside the graphite crucible, insert into the quartz tube, and clamp the KF40 flange finger-tight with the O-ring.
2. **Pump down.** Close the vent valve, start the roughing/turbo pumps, and wait for high vacuum ($\sim 1\times10^{-6}$ Torr or better).
3. **Cooling water.** Open the facility cooling-water valve (~45°, half open) and the generator chiller; check for leaks and confirm flow before applying RF power.
4. **Power on.** Bring up the generator per its sequence, then run the LabVIEW ramp/soak profile (analog setpoint to the generator; pyrometer feedback for closed-loop control).
5. **Anneal.** Hold the temperature/time profile (e.g. ~1200 °C for several hours for Ni grain growth); the VI can email status alerts.
6. **Shutdown.** Ramp power down, stop RF, then turn off cooling water, allow the sample to cool under vacuum, and vent only after the turbo pump has stopped.

Validation and characterization

Thermal performance. The repository contains 100+ logged runs (`docs/data_log/`, IFrun001–100) across Ni200, Ni4N5, YSZ, and Pd thermal-evaporation experiments, with temperatures of

~900–1400 °C and soak durations from minutes to ~40 h, each with `.xlsx`/`.txt`/`.lvm` traces and some photos. Proposed thermal validation figures:

1. **Power-setpoint → temperature calibration** curve (the SOP notes an approximately linear current↔temperature correlation).
2. **Representative ramp/soak profile** (e.g. a 1200 °C multi-hour Ni anneal) parsed from a data log via `plotheatcurve.m` or a Python equivalent.
3. **Control performance**: temperature stability before vs. after adding PID + pyrometer closed-loop control.

Microstructural validation (SEM + EBSD). The intended outcome of the furnace is controlled grain growth, so the key validation is **real microstructural data**:

- **SEM micrographs** of as-received vs. annealed samples (e.g. Ni200 / Ni4N5) showing the increase in grain size produced by the anneal.
- **EBSD orientation and grain-size maps** quantifying grain-size distribution (and, where multiple runs are available, grain size vs. anneal temperature/time), confirming that the closed-loop thermal profile yields the targeted microstructure.

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[TODO: add the SEM and EBSD image files and identify the
corresponding data_log run IDs; source candidate micrographs from
docs/student-work/RyanWeber.pdf and add raw images under paper/
figures/.]
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Safety

- **High voltage / RF.** Lethal voltages and strong RF fields are present near the coil. Keep conductive objects, jewelry, watches, phones, and electronics ≥6–12″ from the coil while powered; nearby conductive hardware can deliver a shock.
- **Lockout-tagout.** Changing the generator’s internal transformer tap (to adjust max power, LEPEL prototype) requires a lockout-tagout procedure.
- **Vacuum.** Never open the manual vent valve while the turbo pump is spinning — this can destroy the pump. Handle the vacuum gauge and quartz tube carefully (both are fragile and costly).
- **Thermal.** Samples reach ~1400–1500 °C; melt/drip and quartz-tube failure are risks. Wear clean nitrile gloves when loading/unloading to avoid contaminating the vacuum.

Ethics statements

This work did not involve human participants, human data, or animal subjects. There are no associated ethical considerations.

Declarations

- **CRediT author roles:** `[TODO]`
- **Acknowledgements:** BYU Mechanical Engineering Department; the Johnson group; Kevin Cole. `[TODO: funding sources.]`
- **Declaration of competing interests:** `[TODO]`
- **Data availability:** design files, logs, and extracted context are in this repository (`vertical-cloud-lab/custom-induction-furnace`).